Ashmit Mukherjee
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Profile

Computer Science student working across computational social science, applied machine learning, and experimental research. I use statistical and computational methods to investigate social, economic, and historical questions, and build systems for data collection, experimentation, and model evaluation.

Education

New York University Abu Dhabi *Expected May 2027*

B.S. in Computer Science. Studied at NYU Abu Dhabi, NYU Paris, and NYU New York.

Relevant coursework: Natural Language Processing; Principles of Data Science; Statistics for Social and Behavioral Sciences; Data Structures and Algorithms; Computer Systems Architecture; Operating Systems.

Experience

Research Assistant, Human-AI Collaboration, New York University Abu Dhabi *2026 to present*

- Lead the technical implementation of an online platform for a human-AI research project.
- Build, deploy, and maintain the system used to run the study.
- Contribute to study design and translate research requirements into platform behavior and data workflows.

Research Assistant, eBRAIN Lab, New York University Abu Dhabi *2026 to present*

- Develop and evaluate parameter-efficient methods for adapting protein language models to protein-sequence tasks.
- Build training and evaluation pipelines and run multi-seed experiments on the NYU Abu Dhabi HPC cluster.
- Compare models and configurations to determine which results hold up across repeated runs.

Machine Learning Engineer Intern, Zeek *June 2025 to August 2025*

- Built and evaluated customer-segmentation workflows using real-world data.
- Compared preprocessing and model choices, then used SHAP to inspect model behavior.
- Contributed to validation and evaluation procedures used in the model-development workflow.

Market Research Consultant, Global Media Alliance Broadcasting *June 2024*

- Led audience analysis, survey design, and customer-engagement analysis for a broadcasting company.
- Synthesized findings into recommendations presented to senior leadership.

Selected Projects

Hinglish Named-Entity Recognition Benchmark

- Fine-tuned mBERT and XLM-RoBERTa on Hindi-English code-mixed text from the COMI-LINGUA dataset.
- Built data-preparation, evaluation, and error-analysis workflows and documented zero-shot baselines.

Salary Data Analysis

- Built a Streamlit project for exploring salary data, comparing models, and inspecting feature contributions with SHAP.
- Used PyCaret and MLflow workflows for model comparison and experiment logging.

CAMP: Campus Asset Management Platform

- Worked on a team-built platform for campus asset reservation and management.
- Contributed to role-based workflows, REST APIs, integration testing, and GitHub Actions CI/CD.

Technical Skills

Languages: Python, C++, C#, JavaScript
Machine learning: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, PyCaret, SHAP
Data and experiments: Pandas, NumPy, SciPy, regression, cross-validation, multi-seed evaluation, experimental design
Systems and tools: FastAPI, Empirica, MLflow, Git, Docker, GitHub Actions, REST APIs, automated testing, Streamlit